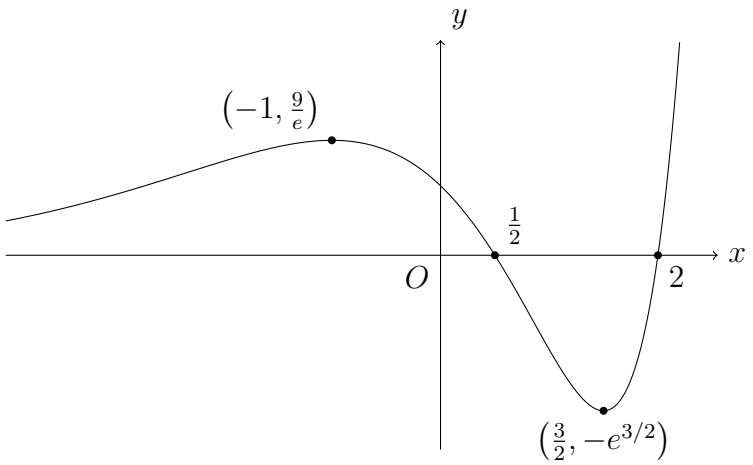
**(i)**

$$y = e^x(2x^2 - 5x + 2) = e^x(2x - 1)(x - 2)$$

$$\frac{dy}{dx} = e^x(2x - 3)(x + 1)$$

$$\frac{dy}{dx} = 0 \implies x = -1,\ \frac{3}{2}$$

| $x$ | $-\infty$ | $-1$ | $\frac{1}{2}$ | $\frac{3}{2}$ | $2$ | $\infty$ |
|---|---|---|---|---|---|---|
| $y$ | $0^+$ | $\frac{9}{e}$ | $0$ | $-e^{\frac{3}{2}}$ | $0$ | $\infty$ |




| $k$ | $k < -e^{\frac{3}{2}}$ | $k = -e^{\frac{3}{2}}$ | $-e^{\frac{3}{2}} < k \leq 0$ | $0 < k < \frac{9}{e}$ | $k = \frac{9}{e}$ | $k > \frac{9}{e}$ |
|---|---|---|---|---|---|---|
| # roots | 0 | 1 | 2 | 3 | 2 | 1 |

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**(ii)**

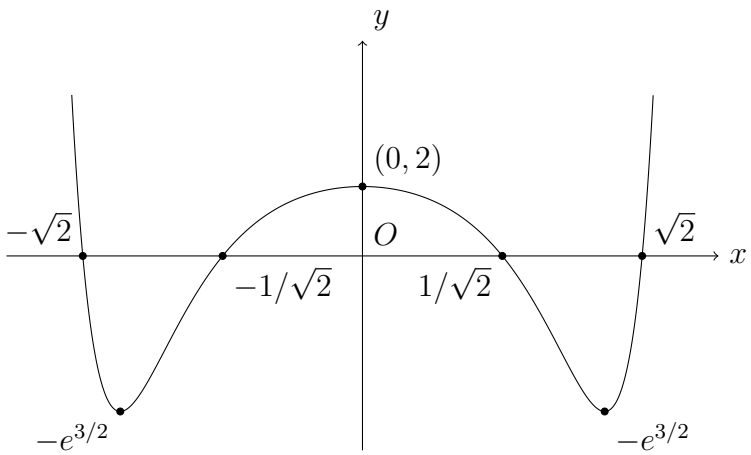
$$f(x) = e^{x^2}(2x^4 - 5x^2 + 2) = e^{x^2}(2x^2 - 1)(x^2 - 2)$$

$$f(x) = f(-x)$$

$$\begin{aligned}\frac{dy}{dx} &= 2xe^{x^2}(2x^4 - 5x^2 + 2) + e^{x^2}(8x^3 - 10x) \\ &= 2xe^{x^2}(2x^2 - 3)(x^2 + 1)\end{aligned}$$

$$\frac{dy}{dx} = 0 \implies x = 0,\ \pm\sqrt{\frac{3}{2}}$$

| $x$ | $0$ | $\frac{1}{\sqrt{2}}$ | $\sqrt{\frac{3}{2}}$ | $\sqrt{2}$ | $\infty$ |
|---|---|---|---|---|---|
| $y$ | $2$ | $0$ | $-e^{\frac{3}{2}}$ | $0$ | $\infty$ |




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This solution was submitted by **Jon**

Published and refined by **Jon**

*Last updated: July 5, 2026*